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B.Tech. Degree V Semester Examination November 2017

ME 15-1503 MACHINE TOOLS AND MACHINING SCIENCE (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What are the differences between automatic lathe and capstan lathe?
 (b) Draw a setup required for cutting external threads on a centre lathe.
 (c) Compare the shaper and planer in terms of operations and types of work pieces.
 (d) Briefly explain differential indexing in a milling machine.
 (e) Define the structure of a grinding wheel.
 (f) What are the basic rules for applying clamping forces in a fixture?
 (g) What are the conditions that would allow a continuous chip to be formed in metal cutting?
 (h) Explain the variables affecting tool life.
 (i) What are the advantages of ceramics as cutting tool materials?
 (j) Define and explain flank wear in machining.

PART B

(4 × 10 = 40)

- II. (a) Explain the characteristic features of the capstan lathe with the help of suitable sketches. (5)
 (b) A work piece with 50 mm diameter is turned in a centre lathe with a back rake angle of -12° and front relief angle of 7° . Calculate the actual cutting angles if the tool is set (i) 1.2 mm above and (ii) 1.2 mm below the centre line of the work piece. (5)

OR

- III. (a) Distinguish between a shaper, planer and a slotter. (5)
 (b) In a steel sheet of 30 mm thickness, 4 holes of 20 mm diameter are to be drilled. The cutting speed is 32 m/min and the feed rate 0.12 mm/revolution. Calculate the machining time and the material removal rate. (5)
- IV. (a) How is the abrasive selected for a grinding operation? Your answer should indicate the reasons for the selection. (5)
 (b) In a slab milling operation, the milling cutter has 20 teeth and a diameter of 100 mm. The speed of the cutter is 5 rotations per second. If the flat surface to be generated is 200 mm × 50 mm and feed per tooth is 0.013 mm/rev, calculate the machining time required for 100 pieces. The depth of cut may be taken as 6 mm. (5)

OR

(P.T.O.)

- V. (a) Explain any three types of locators used in jigs and fixtures. (3)
 (b) Why must the clamps on milling fixtures be extremely rigid? (2)
 (c) Explain the features of a leaf jig with the help of suitable sketches. (5)
- VI. (a) Derive the expression for shear angle in orthogonal cutting in terms of rake angle and chip thickness ratio. (5)
 (b) Explain the conditions which promote the growth of build-up-edge (BUE) in machining with a neat sketch. (5)
- OR**
- VII. (a) Explain the tool geometry of a single point cutting tool with suitable sketches. (5)
 (b) During orthogonal machining with a cutting tool having a 10° rake angle, the chip thickness is measured to be 0.4 mm and the uncut thickness is 0.15 mm. Determine the shear plane angle and also the magnitude of the shear strain. (5)
- VIII. (a) What are the various forms of wear found in cutting tool? Show with neat sketches. (4)
 (b) In an orthogonal cutting experiment with a tool of rake angle 8° , the following observations were made: chip thickness ratio = 0.31, horizontal component of cutting force = 1300 N, vertical component of cutting force = 1600 N. From Merchant's theory, calculate various components of the cutting forces and the coefficient of friction at the chip-tool interface. (6)
- OR**
- IX. (a) Explain the major criteria for judging machinability. (3)
 (b) What are the situations in which diamonds are used as cutting tools? (3)
 (c) Two single point lathe tools made of HSS and carbide have the same tool life of 60 minutes at a cutting speed of 70 m/min. The exponent of tool life in Taylor's equation (n) is 0.12 for HSS while it is 0.21 for carbide. Compare the life of the two tools at a speed of 85 m/min. (4)
